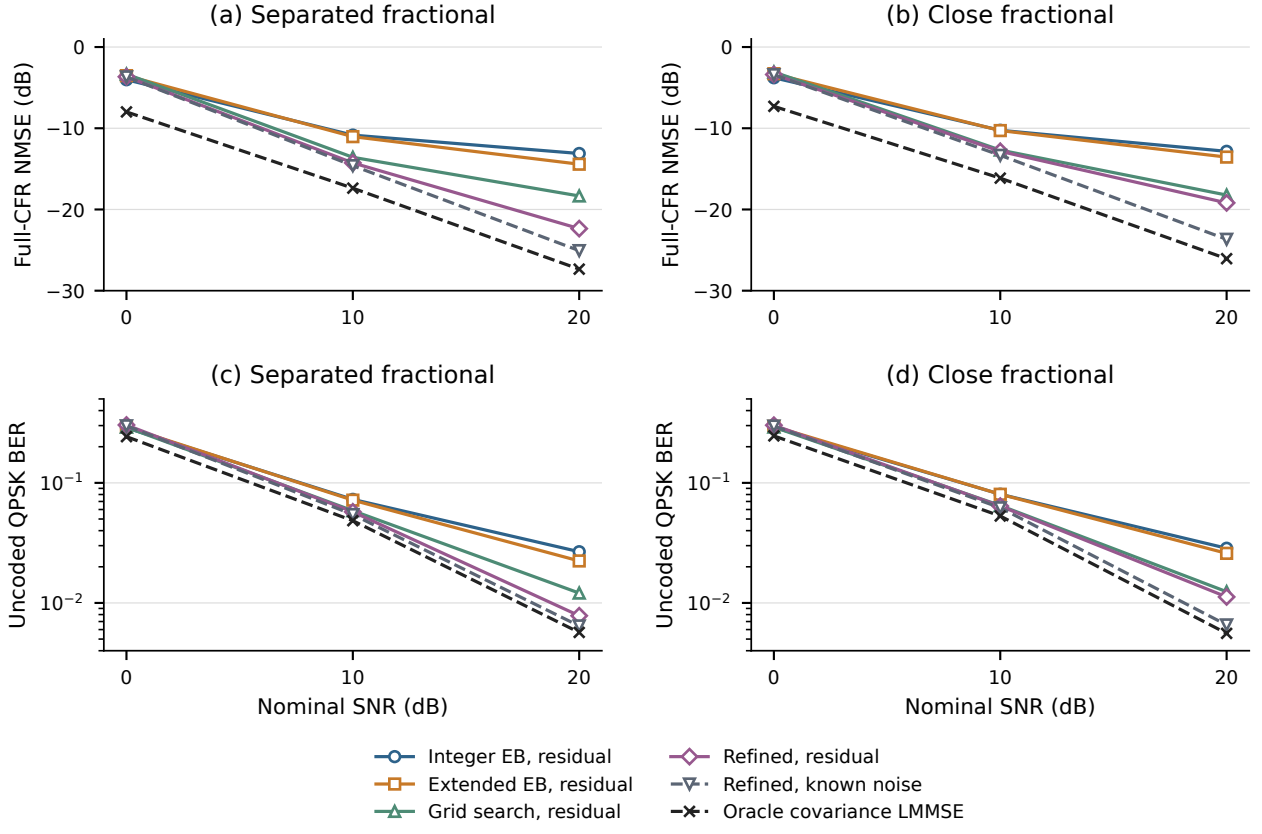
(a) Separated fractional
(b) Close fractional
(c) Separated fractional
(d) Close fractional
Full-CFR NMSE (dB)
Uncoded QPSK BER
Nominal SNR (dB)
0
−10
−20
−30
10
20
10^{-1}
10^{-2}
Integer EB, residual
Extended EB, residual
Grid search, residual
Refined, residual
Refined, known noise
Oracle covariance LMMSE